

CONSERVATION AT EVERY SCALE

From Bookkeeping to the General Transport Equations

A rectangular control-volume derivation using ψ per unit mass and J_ψ as flux

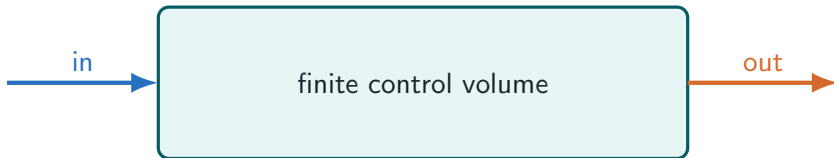
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In – Out + Generation – Consumption = Accumulation

PART I

The Integral Approach

Conservation over a finite control volume



Integral and differential views are the same conservation law

Integral / macroscopic

- ▶ finite region in space;
- ▶ total inventory inside;
- ▶ net transfer across the whole boundary;
- ▶ ideal for tanks, pipes, nozzles and junctions.

$$\frac{d}{dt} \int_{CV} C_\psi dV + \oint_{CS} \mathbf{J}_\psi \cdot \mathbf{n} dA = \int_{CV} S_\psi dV$$

Differential / local

- ▶ shrink the CV toward a point;
- ▶ resolve internal spatial variation;
- ▶ obtain a PDE;
- ▶ ideal for profiles, gradients and fields.

$$\partial_t C_\psi + \nabla \cdot \mathbf{J}_\psi = S_\psi$$

The local equation is the differential limit of the integral balance.

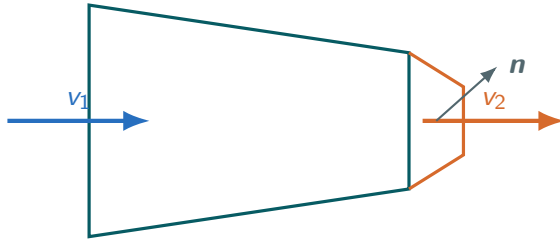
Mass conservation over a finite control volume



$$\oint_{CS} \rho(\mathbf{v} \cdot \mathbf{n}) dA + \frac{\partial}{\partial t} \int_{CV} \rho dV = 0$$

$\mathbf{v} \cdot \mathbf{n} < 0$ at an inlet and > 0 at an outlet. For steady flow the accumulation term vanishes; for steady incompressible 1-D ports, $\sum \dot{m}_{out} = \sum \dot{m}_{in}$.

A slanted control surface: projected area is what matters



The elemental flow is $dQ = (\mathbf{v} \cdot \mathbf{n})dA = v \cos \theta dA$. If a section is cut obliquely, dA increases by $1/\cos \theta$ while the dot product supplies $\cos \theta$. They cancel.

This is why control surfaces need not be perpendicular to the flow.

WELTY 4.4 — converging square channel with an oblique exit

Steady incompressible continuity gives $Q_1 = Q_2$:

$$10 \left(\frac{4}{12} \right)^2 = v_2 \left(\frac{2}{12} \right)^2.$$

Hence

$$\boxed{v_2 = 40 \text{ ft/s}}, \quad \boxed{Q = 1.11 \text{ ft}^3/\text{s}}.$$

The 30° cut does not alter the result: use $\mathbf{v} \cdot \mathbf{n}$ with the actual slanted area, or equivalently use the area projected normal to the velocity.

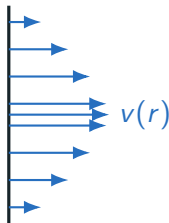
WELTY 4.3 — a nonuniform velocity profile must be integrated

For $v(r) = 9(1 - r^2/16)$ ft/s over a circular region of diameter 1.5 ft, $R = 0.75$ ft:

$$\bar{v} = \frac{1}{\pi R^2} \int_0^R v(r) 2\pi r dr.$$

Thus

$$\bar{v} \simeq 8.84 \text{ ft/s}.$$



The integral weights velocity by annular area $2\pi r dr$.

Wetty 4.22 — uniform inlet, parabolic outlet

The inlet mean velocity is $V_0 = 8 \text{ cm/s}$. Downstream, $v_x = az(z_0 - z)$. Continuity requires $\bar{v}_{out} = 8 \text{ cm/s}$. For this parabola,

$$\bar{v} = \frac{1}{z_0} \int_0^{z_0} az(z_0 - z) dz = \frac{2}{3} v_{max}.$$

Therefore

$$v_{max} = 12 \text{ cm/s}.$$

Continuity constrains the integral of a profile, not its pointwise shape.

Welty 4.16 — filling a V-shaped tank: accumulation survives

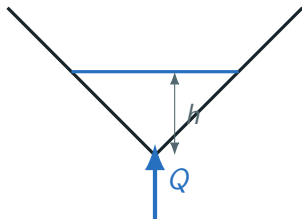
For a tank of width w into the page and half-angle 20° measured from the vertical, the liquid volume is

$$V(h) = wh^2 \tan 20^\circ.$$

The integral balance is simply $dV/dt = Q$, hence

$$\frac{dh}{dt} = \frac{Q}{2wh \tan 20^\circ},$$

$$t_{1 \rightarrow 2} = \frac{w \tan 20^\circ}{Q} (h_2^2 - h_1^2).$$



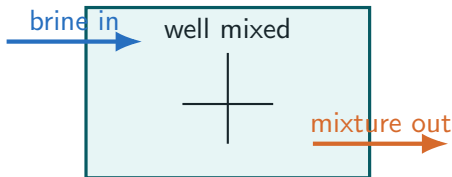
Welty 4.7 — mixing tank: total mass first, species mass second

A well-mixed tank is the clearest bridge from integral bookkeeping to an ODE. Write two balances:

$$\frac{dM}{dt} = \dot{m}_{in} - \dot{m}_{out}$$

and, for salt mass S ,

$$\frac{dS}{dt} = w_{in}\dot{m}_{in} - \frac{S}{M(t)}\dot{m}_{out}.$$



The outlet composition is the instantaneous tank composition because of the well-mixed assumption.

PART II

The Differential Approach

Shrink the control volume and resolve the field

Start with a specific transported quantity

Let

$$\psi = \frac{\text{amount of transported property}}{\text{mass}}.$$

Then the corresponding amount per unit volume is

$$\boxed{\rho\psi}.$$

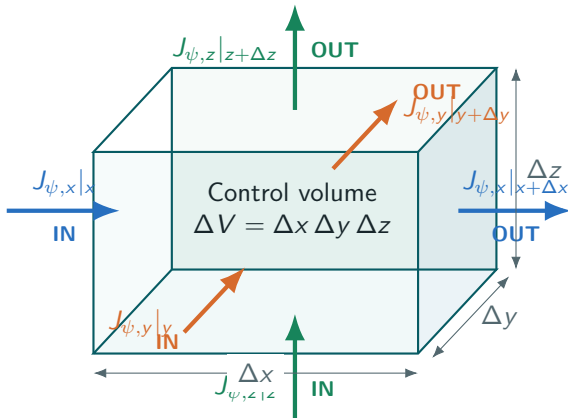
For a small volume ΔV , the amount stored is

$$\boxed{\text{stored amount} = (\rho\psi)\Delta V}.$$

Process	ψ	$\rho\psi$
Total mass	1	ρ
Species A	w_A	ρw_A
Energy	e	ρe
x-momentum	u_x	ρu_x

ψ is the transported property per unit mass; $\rho\psi$ is the corresponding amount per unit volume.

2. Choose a rectangular parallelepiped control volume



Six faces occur in three opposing pairs:

- ▶ two normal to x ,
- ▶ two normal to y ,
- ▶ two normal to z .

Observe the system for a finite time interval Δt .

3. The bookkeeping statement is exact before any calculus

For the finite control volume during Δt ,

$$\text{In} - \text{Out} + \text{Generation} - \text{Consumption} = \text{Accumulation}$$

Each term is an **amount**, not a rate:

amount crossing a face = (flux)(face area)(Δt),

amount produced internally = (volumetric rate)(ΔV)(Δt),

accumulation = (final stored amount) – (initial stored amount).

Dimensional checking is the first defence against a wrong balance.

4. Write the contribution of the two x -faces

The area normal to x is $\Delta A_x = \Delta y \Delta z$.

$$\text{In through } x = J_{\psi,x}|_x \Delta y \Delta z \Delta t,$$

$$\text{Out through } x + \Delta x = J_{\psi,x}|_{x+\Delta x} \Delta y \Delta z \Delta t.$$

Therefore,

$$(\text{In} - \text{Out})_x = [J_{\psi,x}|_x - J_{\psi,x}|_{x+\Delta x}] \Delta y \Delta z \Delta t.$$

The ordering **in minus out** will later generate a negative spatial derivative.

5. Repeat the same argument for y and z

$$(\text{In} - \text{Out})_y = [J_{\psi,y}|_y - J_{\psi,y}|_{y+\Delta y}] \Delta x \Delta z \Delta t,$$

$$(\text{In} - \text{Out})_z = [J_{\psi,z}|_z - J_{\psi,z}|_{z+\Delta z}] \Delta x \Delta y \Delta t.$$

The total net influx is the sum over all three face pairs:

$$(\text{In} - \text{Out})_{\text{faces}} = (\text{In} - \text{Out})_x + (\text{In} - \text{Out})_y + (\text{In} - \text{Out})_z.$$

A positive $J_{\psi,x}$ is defined in the positive x direction; the control-volume signs follow automatically.

6. Add internal sources and the change in storage

Let $\dot{r}_{\psi}^{+}$ and $\dot{r}_{\psi}^{-}$ be the volumetric generation and consumption rates. Then

$$\text{Generation} - \text{Consumption} = \underbrace{(\dot{r}_{\psi}^{+} - \dot{r}_{\psi}^{-})}_{S_{\psi}} \Delta V \Delta t.$$

The stored amount changes from $(\rho\psi)_t \Delta V$ to $(\rho\psi)_{t+\Delta t} \Delta V$. Therefore,

$$\text{Accumulation} = [(\rho\psi)_{t+\Delta t} - (\rho\psi)_t] \Delta V.$$

Generation and consumption act throughout the volume; accumulation compares two times. No steady-state assumption has been made.

7. Assemble the complete finite-volume balance

$$\begin{aligned} & [J_{\psi,x}|_x - J_{\psi,x}|_{x+\Delta x}] \Delta y \Delta z \Delta t \\ & + [J_{\psi,y}|_y - J_{\psi,y}|_{y+\Delta y}] \Delta x \Delta z \Delta t \\ & + [J_{\psi,z}|_z - J_{\psi,z}|_{z+\Delta z}] \Delta x \Delta y \Delta t \\ & + \left(\dot{r}_{\psi}^+ - \dot{r}_{\psi}^- \right) \Delta x \Delta y \Delta z \Delta t \\ & = [(\rho\psi)_{t+\Delta t} - (\rho\psi)_t] \Delta x \Delta y \Delta z. \end{aligned}$$

This algebraic equation is already a valid discrete control-volume model.

8. Divide by volume and time to obtain rate densities

Divide the finite balance by $\Delta V \Delta t$:

$$\begin{aligned} & - \frac{J_{\psi,x}|_{x+\Delta x} - J_{\psi,x}|_x}{\Delta x} - \frac{J_{\psi,y}|_{y+\Delta y} - J_{\psi,y}|_y}{\Delta y} \\ & - \frac{J_{\psi,z}|_{z+\Delta z} - J_{\psi,z}|_z}{\Delta z} + S_\psi = \frac{(\rho\psi)_{t+\Delta t} - (\rho\psi)_t}{\Delta t}. \end{aligned}$$

Now every term has units

$$\frac{\text{amount of } \psi}{\text{volume} \times \text{time}}.$$

The minus signs arise from **in minus out**, not from a constitutive law.

9. Shrinking the control volume gives the local equation

Take

$$\Delta x, \Delta y, \Delta z, \Delta t \rightarrow 0.$$

Then

$$\frac{\partial(\rho\psi)}{\partial t} = -\frac{\partial J_{\psi,x}}{\partial x} - \frac{\partial J_{\psi,y}}{\partial y} - \frac{\partial J_{\psi,z}}{\partial z} + S_{\psi}.$$

Using $\mathbf{J}_{\psi} = (J_{\psi,x}, J_{\psi,y}, J_{\psi,z})$,

$$\frac{\partial(\rho\psi)}{\partial t} + \nabla \cdot \mathbf{J}_{\psi} = S_{\psi}.$$

This is the general local conservation equation.

10. Total flux contains bulk and molecular transport

The total flux is decomposed as

$$\mathbf{J}_\psi = \rho \mathbf{u} \psi + \mathbf{J}_\psi^{(\text{mol})},$$

where

- ▶ $\rho \mathbf{u} \psi$ is convective transport by bulk motion;
- ▶ $\mathbf{J}_\psi^{(\text{mol})}$ is molecular transport relative to that motion.

Substitution gives

$$\frac{\partial(\rho\psi)}{\partial t} + \nabla \cdot (\rho \mathbf{u} \psi) = -\nabla \cdot \mathbf{J}_\psi^{(\text{mol})} + \mathcal{S}_\psi.$$

Conservation determines the structure; a constitutive law supplies $\mathbf{J}_\psi^{(\text{mol})}$.

11. Total-mass conservation gives the continuity equation

For total mass,

$$\psi = 1, \quad \mathbf{J}_1^{(\text{mol})} = \mathbf{0}, \quad S_\psi = 0.$$

Therefore,

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0.$$

For constant density,

$$\nabla \cdot \mathbf{u} = 0.$$

For one-dimensional constant-area flow,

$$\frac{du_x}{dx} = 0.$$

Continuity is not an optional constitutive relation; it is the total-mass balance.

12. Distinguish the specific field from its conserved density

The transported property per unit mass is ψ . Its conserved amount per unit volume is

$$C_\psi \equiv \rho\psi.$$

These are different fields whenever density varies:

$$\nabla C_\psi = \nabla(\rho\psi) = \rho\nabla\psi + \psi\nabla\rho.$$

Specific-field gradient

$$\nabla\psi$$

Ignores density variation.

Conserved-density gradient

$$\nabla(\rho\psi)$$

Contains both $\nabla\psi$ and $\nabla\rho$.

13. A unified density-diffusion closure can be postulated

If diffusion is assumed to be driven by the conserved-density gradient, write

$$\mathbf{J}_{\psi}^{(\text{mol})} = -\Gamma_{\psi} \nabla(\rho\psi).$$

Its dimensions are correct:

$$[\Gamma_{\psi} \nabla(\rho\psi)] = \frac{\text{m}^2}{\text{s}} \frac{[\rho\psi]}{\text{m}} = \frac{\text{property}}{\text{m}^2 \text{s}}.$$

The governing equation becomes

$$\frac{\partial(\rho\psi)}{\partial t} + \nabla \cdot (\rho \mathbf{u} \psi) = \nabla \cdot [\Gamma_{\psi} \nabla(\rho\psi)] + S_{\psi}.$$

This closure allows $\rho = \rho(\mathbf{x}, t)$; no constant-density assumption is needed to write it.

14. Another common closure diffuses the specific field

Many standard constitutive laws instead have the form

$$\mathbf{J}_{\psi}^{(\text{mol})} = -\rho \Gamma_{\psi} \nabla \psi.$$

Compare the two choices:

$$-\Gamma_{\psi} \nabla(\rho\psi) = -\rho \Gamma_{\psi} \nabla \psi - \Gamma_{\psi} \psi \nabla \rho,$$

$$-\rho \Gamma_{\psi} \nabla \psi = \text{specific-field-gradient closure.}$$

They coincide when

$$\psi \nabla \rho = \mathbf{0}.$$

Conservation is universal; the choice of driving gradient is constitutive physics.

15. Heat: enthalpy density and Fourier conduction

Take $\psi = h$, with $h = c_p T$ for a calorically simple material. The conserved density is

$$C_h = \rho h = \rho c_p T.$$

The density-diffusion proposal is

$$\mathbf{J}_h^{(\text{mol})} = -\alpha \nabla(\rho h).$$

Fourier's law is

$$\mathbf{J}_h^{(\text{mol})} \equiv \mathbf{q} = -k \nabla T.$$

If ρc_p is spatially uniform and $\alpha = k/(\rho c_p)$, then

$$-\alpha \nabla(\rho h) = -\alpha \rho c_p \nabla T = -k \nabla T = \mathbf{q}.$$

Under those conditions, the two expressions are identical.

16. Variable properties expose the limitation of the shortcut

In general,

$$\begin{aligned} -\alpha \nabla(\rho h) &= -\alpha \nabla(\rho c_p T) \\ &= -\alpha \rho c_p \nabla T - \alpha c_p T \nabla \rho - \alpha \rho T \nabla c_p. \end{aligned}$$

Fourier's law contains only

$$-k \nabla T.$$

Therefore,

$$-\alpha \nabla(\rho h) \neq -k \nabla T$$

in general variable-density, variable- c_p media.

Thermal diffusivity unifies the PDE after assumptions; temperature remains the Fourier driving field.

17. Species: concentration and mass-fraction forms differ

For species A , let $\psi = w_A$. Then

$$C_A = \rho w_A = \rho_A.$$

Diffusion of species concentration gives

$$\mathbf{J}_A^{(\text{mol})} = -D_A \nabla \rho_A = -D_A \nabla (\rho w_A).$$

The barycentric mass-fraction form is commonly written

$$\mathbf{J}_A^{(\text{mol})} = -\rho D_A \nabla w_A.$$

Their difference is

$$-D_A w_A \nabla \rho.$$

The appropriate form depends on concentration variable, reference velocity and mixture model.

18. Momentum: viscosity diffuses velocity, not generally momentum density

For x -momentum, $\psi = u_x$ and $C_{p_x} = \rho u_x$. A density-diffusion model gives

$$J_{p_x, y}^{(\text{mol})} = -\nu \frac{\partial(\rho u_x)}{\partial y}.$$

Expanding,

$$J_{p_x, y}^{(\text{mol})} = -\rho \nu \frac{\partial u_x}{\partial y} - \nu u_x \frac{\partial \rho}{\partial y}.$$

Newtonian momentum transport is

$$J_{p_x, y}^{(\text{mol})} = -\mu \frac{\partial u_x}{\partial y}, \quad \mu = \rho \nu.$$

The extra density-gradient term is not part of the ordinary Newtonian constitutive law.

19. The three diffusivities share one dimensional scale

Process	Field	Diffusivity	Standard simple closure
Heat	T	$\alpha = \frac{k}{\rho c_p}$	$\mathbf{J}_h^{(\text{mol})} = -k \nabla T$
Species	w_A	D_A	$\mathbf{J}_A^{(\text{mol})} = -\rho D_A \nabla w_A$
Momentum	u_x	$\nu = \frac{\mu}{\rho}$	$J_{p_x, y}^{(\text{mol})} = -\mu \frac{\partial u_x}{\partial y}$

$$[\alpha] = [D_A] = [\nu] = \text{m}^2 \text{s}^{-1}.$$

They quantify the rate at which thermal, compositional and momentum disturbances spread.

20. Under simple conditions the same diffusion equation emerges

For a stationary medium with constant properties and no source,

$$\frac{\partial \phi}{\partial t} = \Gamma \nabla^2 \phi.$$

Heat

$$\frac{\partial T}{\partial t} = \alpha \nabla^2 T$$

Species

$$\frac{\partial w_A}{\partial t} = D_A \nabla^2 w_A$$

Momentum

$$\frac{\partial u_x}{\partial t} = \nu \nabla^2 u_x$$

The common characteristic time is

$$t_{\text{diff}} \sim \frac{L^2}{\Gamma}.$$

The analogy is exact at the PDE level only after the stated simplifications.

21. Ratios of diffusivities predict competing transport rates

Because all three coefficients have the same dimensions, their ratios are dimensionless:

$$\begin{aligned} \text{Pr} &= \frac{\nu}{\alpha} && \text{momentum diffusion / thermal diffusion,} \\ \text{Sc} &= \frac{\nu}{D_A} && \text{momentum diffusion / mass diffusion,} \\ \text{Le} &= \frac{\alpha}{D_A} && \text{thermal diffusion / mass diffusion.} \end{aligned}$$

For example,

$$\text{Pr} > 1 \implies \text{momentum spreads faster than temperature over the same } L.$$

These ratios connect the mathematical analogy to boundary-layer thicknesses and time scales.

22. A deeper closure uses the true thermodynamic driving force

The safest general constitutive statement is

$$\mathbf{J}_{\psi}^{(\text{mol})} = -L_{\psi} \nabla \Phi_{\psi},$$

where Φ_{ψ} is the appropriate driving potential.

- ▶ Heat conduction: temperature or reciprocal-temperature driving force.
- ▶ Species diffusion: chemical-potential gradients, with Fick's law as a simplification.
- ▶ Momentum transport: velocity-gradient tensor through the viscous stress law.

The minus sign expresses relaxation toward uniform driving potential; the potential itself is process dependent.

23. The unified hierarchy prevents conceptual shortcuts

1. **Define storage:** $C_\psi = \rho\psi$.
2. **Apply conservation:** $\partial_t C_\psi + \nabla \cdot \mathbf{J}_\psi = S_\psi$.
3. **Separate transport:** $\mathbf{J}_\psi = \rho \mathbf{u} \psi + \mathbf{J}_\psi^{(\text{mol})}$.
4. **Choose the physical closure:** $\mathbf{J}_\psi^{(\text{mol})} = -L_\psi \nabla \Phi_\psi$.
5. **Introduce simplifying assumptions** before identifying α , D_A or ν in a common diffusion PDE.

Never infer the constitutive law from conservation alone.

24. The derivation reveals what a global balance misses

Macroscopic balance

- ▶ tracks total inventory;
- ▶ measures net boundary transfer;
- ▶ includes overall production or consumption;
- ▶ cannot resolve internal gradients.

Local transport equation

- ▶ resolves $\rho\psi(\mathbf{x}, t)$;
- ▶ distinguishes individual face fluxes;
- ▶ exposes gradients and hot spots;
- ▶ predicts the evolving field.

$$\text{In} - \text{Out} + \text{Generation} - \text{Consumption} = \text{Accumulation}$$

The PDE is conservation bookkeeping in the differential limit.

References for further study

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The notation in this deck deliberately separates universal conservation from process-specific constitutive closure.